Path Ramsey Thresholds in Coprime Graphs

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Abstract

Let G_n be the graph on $\{1, \ldots, n\}$ whose edges join coprime integers, and let $P_{\mathrm{cop}}(k)$ be the least n such that every two-coloring of its edges contains a monochromatic simple path with k edges. We prove $P_{\mathrm{cop}}(9) = 17$. The upper bound combines explicitly checked finite path certificates with a classification: every connected 17-vertex graph with 48 edges and no ten-vertex path has a vertex cover of size at most four or is isomorphic to $K_3 \vee (K_3 \cup \overline{K}_{11})$. Both possibilities contradict a deletion condition forced by the certificates. We also give the bounds $2 \leqslant \liminf_{k\to\infty} P_{\mathrm{cop}}(k)/k \leqslant \limsup_{k\to\infty} P_{\mathrm{cop}}(k)/k \leqslant 197\pi^2/595$, and describe independently checkable certificates for the previously reported values $P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13$.

Mathematics Subject Classifications: 05C38, 05C55, 05D10, 11A05

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1 Introduction

The coprime graph G_n has vertex set $[n] = \{1, \ldots, n\}$ and edge set $\{\{a, b\} : 1 \leqslant a < b \leqslant n,\ \gcd(a, b) = 1\}$. We study its two-color edge Ramsey threshold for a simple path. Throughout, $P_{\mathrm{cop}}(k)$ counts *edges* in the target, while P_j denotes a path on j *vertices*. Thus a coloring counted as avoiding at parameter k has two P_{k+1}-free color graphs. Color graphs have the full vertex set, including any isolated vertices. The notation $X \vee Y$ denotes the join of disjoint graphs, and $\overline{K}_t$ is an independent set of order t.

The clique thresholds in the same host have a different explanation. Du, Xi, Deng, and Ma [4] transfer complete edge targets to classical Ramsey numbers through prime labels. Their Table 12 cites earlier path computations in this repository, and their Open Problem 9.2 asks about non-complete targets. A label pullback need not preserve distinct vertices of a path. The present paper addresses one exact path threshold and general linear bounds.

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Theorem 1. *Every two-coloring of $E(G_{17})$ contains a monochromatic P_{10}, whereas some two-coloring of $E(G_{16})$ contains none. Consequently, $P_{\mathrm{cop}}(9) = 17$.*

The finite part of the upper bound verifies paths for specified cuts and deletions of G_{17}; it does not enumerate all colorings. It reduces a hypothetical avoiding coloring to two connected color graphs with 47 and 48 edges. The following graph-theoretic statement then supplies the contradiction.

Theorem 2. *Let J be a connected P_{10}-free graph on 17 vertices with 48 edges. Either J has a vertex cover of size at most four, or*

$$J \cong K_3 \vee (K_3 \cup \overline{K}_{11}).$$

In either case, deleting some four vertices leaves at most one edge.

The proof uses stability results of Füredi, Kostochka, and Verstraëte [6] and Füredi, Kostochka, Luo, and Verstraëte [7], a longest-path classification of Ali and Staton [1], and Hamiltonian closure. The edge count lies on a strict stability threshold, so we give the equality analysis in full. We make no assertion that this classification is absent from all prior literature. The comparison with the later refinements of Ma and Ning [8] is discussed in Section 6.

Theorem 3. *For $k \geqslant 3$,*

$$P_{\mathrm{cop}}(k) \geqslant 4 \left\lfloor \frac{k-1}{2} \right\rfloor + 1,$$

and

$$2 \leqslant \liminf_{k\to\infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \limsup_{k\to\infty} \frac{P_{\mathrm{cop}}(k)}{k} \leqslant \frac{197\pi^2}{595} = 3.267751373\ldots.$$

In particular, $P_{\mathrm{cop}}(k) = \Theta(k)$.

The linear order follows from standard extremal and arithmetic estimates. The displayed upper constant comes from deleting an explicit independent set before applying the density bound. Neither optimality nor priority of this refinement is asserted here.

2 Three finite properties of the host

Write $G[A, B]$ for the graph of host edges across disjoint vertex sets A, B. The following lemma consists of finite statements with positive path certificates. Their files and independent checking procedure are specified in Appendix A.

Lemma 4. *The graph G_{17} has the following properties.*

1. *If $A \cup B = [17]$ is a partition with $|A|, |B| \geqslant 5$, then $G_{17}[A, B]$ contains a P_{10}.*
2. *For every four-element set $A \subset [17]$ and every set F of at most four edges of $G_{17} - A$, the graph $G_{17} - A - F$ contains a P_{10}.*

3. *If* $|A| = 4$, $e(G_{17}[A]) \geqslant 5$, *and* $B = [17] \setminus A$, *then for every edge* $e \in E(G_{17}[B])$, *the graph with edge set* $E(G_{17}[A, B]) \cup \{e\}$ *contains a* P_{10}.

Finite certificate verification. For part 1, the archive gives a crossing path for each of the $\sum_{s=5}^{8} \binom{17}{s} = 62\,322$ smaller sides. Taking complements covers the remaining orientations. For part 3, it gives one path for every qualifying pair (A, e): 63 859 pairs on 1 253 sets A.

For part 2, there are $\binom{17}{4} = 2380$ vertex deletions. Verified host automorphisms reduce these to 262 deletion orbits. For each orbit, the archive gives a family of surviving paths with no edge transversal of size at most four; the families contain 30 671 paths in total. A transversal is a set meeting every path's edge set. Its nonexistence is checked by exhaustive branching: choose an unhit path and branch on each of its edges as the next element of the transversal, rejecting branches that use more than four edges. Every possible transversal is covered by this recurrence.

The checker derives the required cases from the task parameters. For every path it verifies ten distinct vertices, all nine gcd adjacencies, and the relevant cut or deletion restrictions. It also checks every transposition within each declared vertex class against host adjacency. These transpositions generate the within-class permutations, so deletions with the same class counts are equivalent. Every possible deletion signature must occur exactly once. Thus the checked families imply the three statements; saved completion flags or search outcomes are not proof premises. □

We use two published connected-path bounds. For a connected 13-vertex P_{10}-free graph, Balister, Győri, Lehel, and Schelp [2, Theorem 1.3] give

$$e \leqslant \max\left\{\binom{8}{2} + 5, \binom{5}{2} + 4 \cdot 8\right\} = 42. \tag{1}$$

For order $s \geqslant 14$, the path stability theorem [6, Theorem 1.6] gives $e \leqslant 3s - 3$ unless the graph is a subgraph of $H(s, 10, 4)$. In general, $H(n, k, a)$ has parts A, B, C of sizes $a, n - k + a, k - 2a$, a clique on $A \cup C$, and all A–B edges. Its edge count is

$$h(n, k, a) = \binom{k - a}{2} + a(n - k + a). \tag{2}$$

Deleting A from $H(s, 10, 4)$ leaves one edge and isolated vertices.

Proposition 5. *If a two-coloring of* G_{17} *has no monochromatic* P_{10}, *both color graphs are connected on all 17 vertices, their edge counts are 47 and 48, and each retains at least five edges after every four-vertex deletion.*

Proof. If all components of one color had order at most 12, a union of its components would have order between 5 and 12. Indeed, use one such component if possible; otherwise accumulate components of order at most four until their union first reaches five, when its order is at most eight. All crossing host edges have the other color, contradicting Lemma 4, part 1. Thus each color has a component of order $s \geqslant 13$.

For $s \geqslant 14$, this component cannot be contained in $H(s, 10, 4)$. Deleting its four distinguished vertices would leave at most $1 + \binom{17-s}{2} \leqslant 4$ edges of its color. Removing those edges and applying Lemma 4, part 2, gives a forbidden path of the other color. Equation (1) and the path stability bound now give the following bounds for the entire color graph, including edges outside its large component:

s	13	14	15	16	17
e	48	42	43	45	48.

Here the first entry is $42 + \binom{4}{2}$, and the others are $3s - 3 + \binom{17-s}{2}$. Since $e(G_{17}) = 95$, the two colors have 47 and 48 edges, and their large components have order 13 or 17.

Suppose one component has vertex set B of size 13, with complement A. At least $47 - 42 = 5$ edges of that color lie in A, so $e(G_{17}[A]) \geqslant 5$. All crossing edges have the other color. Lemma 4, part 3, forces every host edge inside B to have the component's color. But Lemma 4, part 2, with no edge deletion, gives a P_{10} in $G_{17}[B]$, a contradiction. Both color graphs therefore span connectedly.

Finally, if deleting four vertices left at most four edges of one color, remove those edges and apply Lemma 4, part 2, once more. This proves the deletion condition. □

3 The extremal equality case

We prove Theorem 2. Adding a universal vertex u to a connected P_{10}-free graph gives a 2-connected graph with no cycle of length at least 11. A cycle through u of that length would leave a P_{10} after removing u, and a cycle avoiding u would itself contain such a path.

Let $\mathcal{F}_0$ be the family of graphs obtained from $K_{5,6}$ by deleting at most two edges; call its two parts A and B, of sizes five and six. We record the stability inputs in the form needed below. For a 2-connected graph Q with $c(Q) < 11$:

1. For $n = |Q| \geqslant 15$, $e(Q) > h(n, 11, 4)$ implies $Q \subseteq H(n, 11, 5)$ [6, Theorem 1.4].
2. For $n \geqslant 11$, $e(Q) > \max\{h(n, 11, 4), h(n, 11, 3)\}$ implies $Q \subseteq H(n, 11, 5)$ or $Q \subseteq H(n, 11, 2)$ [7, Theorem 2.3].
3. For $n \geqslant 11$, a subgraph of $H(n, 11, 5)$ with more than $h(n, 11, 4)$ edges contains a member of $\mathcal{F}_0$ [7, Lemma 4.6].
4. If Q/xy is 2-connected and contains $\mathcal{F}_0$, then Q contains $\mathcal{F}_0$ [6, Lemma 4.9].

Here contraction suppresses loops and parallel edges, and

$$h(n, 11, 4) = 4n - 7, \qquad h(n, 11, 3) = 3n + 4, \qquad h(n, 11, 2) = 2n + 18. \tag{3}$$

Every pair of distinct vertices of a member of $\mathcal{F}_0$ is joined within it by a path of at least eight edges; if the pair is not entirely in A, there is one of at least nine edges [6, Lemma 4.8]. In particular, a B–B pair has a ten-edge path by parity.

3.1 The dense bipartite configuration

Lemma 6. *If a 2-connected graph Q has circumference less than 11 and contains a member of $\mathcal{F}_0$ with parts A, B, then $Q - A$ is independent. If $Q = J + u$ with u universal and $|J| \geqslant 10$, then J has a vertex cover of size four.*

Proof. Each component S outside $A \cup B$ has at least two attachments there. If $|S| \geqslant 2$, the attachment edges meet at least two vertices on each side: a unique incident vertex on either side would be a cut vertex. Choose attachment edges ax, by with $a \neq b$ in $A \cup B$ and $x \neq y$ in S. A path in S from x to y gives an a–b path of at least three edges, internally outside $A \cup B$. Closing it with an eight-edge path in $\mathcal{F}_0$ gives a cycle of length at least 11. Hence every S is a singleton.

If such a singleton had an attachment in B, a second attachment and a nine-edge path between the attachments in $\mathcal{F}_0$ would give the same contradiction. Thus all outside vertices attach only to A. An edge inside B would close a ten-edge path in $\mathcal{F}_0$, so there is none. This proves that $Q - A$ is independent. In the universal augmentation, u belongs to A, since vertices outside A have degree at most five. Removing u leaves a four-vertex cover of J. □

Assume henceforth that the graph J in Theorem 2 has no four-vertex cover. Its universal augmentation contains no member of $\mathcal{F}_0$, and neither does any induced augmentation used below.

3.2 Reduction to ten vertices

The augmentation of J has 18 vertices and 65 edges. If an edge xy of J belongs to at most one triangle, it belongs to at most two triangles in the augmentation. Contracting it therefore leaves at least 62 edges on 17 vertices. The contracted graph is still a universal augmentation of a connected graph, hence is 2-connected; contraction cannot increase circumference. Since $62 > h(17, 11, 4) = 61$, the stability and lifting statements force $\mathcal{F}_0$ before contraction, a contradiction.

We now maintain the following conditions for a current induced graph L of order n:

$$L \text{ is connected and } P_{10}\text{-free}, \quad e(L) = 3n - 3, \quad \delta(L) \geqslant 2, \quad L + u \text{ contains no } \mathcal{F}_0. \quad (4)$$

Initially they hold, and the preceding argument even gives $\delta(J) \geqslant 3$. Suppose $n \geqslant 11$. A contraction of an edge in at most one triangle of L would leave an n-vertex augmentation with at least $4n - 6$ edges. This exceeds both $4n - 7$ and $3n + 4$. For $n \geqslant 13$, it also exceeds $h(n, 11, 2) = 2n + 18$, so the smaller-order stability theorem again forces $\mathcal{F}_0$.

For $n = 11, 12$, the $H(n, 11, 2)$ alternative requires a further argument. Contracting an edge in at most one triangle in a connected graph of minimum degree at least two and order greater than three creates at most one leaf. An old vertex can become a leaf only if it is the unique common neighbor of the endpoints. The merged vertex can become a leaf only if both endpoints have degree two with that same neighbor; both events together

would make an isolated triangle. But the universal vertex in a subgraph of $H(n, 11, 2)$ lies in its two-vertex joining part. Removing it leaves a connected subgraph of

$$K_1 \vee (K_7 \cup \overline{K}_{n-9}),$$

which has at least $n - 9 \geqslant 2$ leaves. This is impossible. Every edge of L therefore lies in at least two triangles, and $\delta(L) \geqslant 3$.

We next rule out $\delta(L) \geqslant 4$. A longest nonspanning path in a connected graph of minimum degree δ has at least $2\delta + 1$ vertices. To see this, let its order be $r \leqslant 2\delta$. Both endpoints have all neighbors on the path. Their neighbor-index sets among its $r - 1$ consecutive positions overlap, closing the path into a cycle. Connectivity then extends it using a vertex outside the cycle, a contradiction. Since $r \leqslant 9 < n$, minimum degree at least four forces $\delta = 4$ and $r = 9$.

Ali and Staton [1, Theorem 3] classify connected graphs with minimum degree $\delta \geqslant 2$, longest path order $2\delta + 1$, and order at least $2\delta + 2$: they are subgraphs of $K_\delta \vee \overline{K}_{n-\delta}$ or windmills $K_1 \vee (tK_\delta)$. Here the windmill has $5(n-1)/2 < 3n-3$ edges. In the other case, minimum degree four forces every cross edge of $K_4 \vee \overline{K}_{n-4}$ to be present. Its augmentation contains $K_{5,6}$, again impossible. We have proved $\delta(L) = 3$.

The three neighbors of a degree-three vertex form a triangle, since each incident edge lies in two triangles. Deleting this vertex preserves connectivity, reduces the edge count by three, and leaves minimum degree at least two. Thus it preserves (4). Seven such deletions give a connected P_{10}-free graph L_{10} on ten vertices with 27 edges and minimum degree at least two. Its predecessor L_{11} had minimum degree at least three. Reverse reconstruction adds vertices whose neighborhoods among the vertices already present are cliques of size exactly three.

3.3 The ten-vertex core

Let $Q = L_{10} + u$. It has 11 vertices, 37 edges, minimum degree at least three, and no Hamiltonian cycle. Take its 11-closure: repeatedly add an edge between nonadjacent vertices whose degree sum is at least 11. The Bondy–Chvátal theorem [3] preserves non-Hamiltonicity.

If any edge is added, the closure has at least 38 edges and still no cycle of length at least 11. Since both stability thresholds in (3) equal 37 at $n = 11$, it is a subgraph of $H(11, 11, 5)$ or $H(11, 11, 2)$. The latter has two vertices of degree two and is excluded by minimum degree. Consequently $L_{10} \subseteq K_4 \vee \overline{K}_6$.

Suppose instead that Q is already closed. Chvátal's degree-sequence criterion (as stated in [7, Theorem 4.2]) gives an index $i < 11/2$ with $d_i \leqslant i$ and $d_{11-i} \leqslant 10 - i$. Minimum degree three restricts i to $3, 4, 5$. Removing u gives the following possibilities.

If $i = 3$, the ascending degrees of L_{10} are bounded termwise by

$$(2, 2, 2, 6, 6, 6, 6, 6, 9, 9).$$

Their sum is $54 = 2e(L_{10})$, so equality holds throughout, forcing $L_{10} = K_2 \vee (K_5 \cup \overline{K}_3)$. For $i = 4$, the corresponding upper sequence is

$$(3, 3, 3, 3, 5, 5, 5, 9, 9, 9),$$

again with sum 54, forcing $L_{10} = K_3 \vee (K_3 \cup \overline{K}_4)$.

For $i = 5$, choose six vertices of degree at most five in Q as B, and let A be the remaining five. Their total degree is 74, just six below $6 \cdot 5 + 5 \cdot 10 = 80$. Two nonadjacent vertices of A would have degree sum at most ten by closedness, creating deficit at least ten. Hence A is a clique.

Put $b = e(Q[B])$, and let m count absent A–B edges. The edge count gives $m = b + 3$. For $v \in B$, let s_v count missing cross edges and t_v count neighbors inside B. Then

$$q_v = s_v - t_v \geqslant 0, \qquad \sum_{v \in B} q_v = 3 - b, \qquad 0 \leqslant b \leqslant 3.$$

If r_a counts missing cross edges at $a \in A$, every absent pair av satisfies

$$(10 - r_a) + (5 - q_v) \leqslant 10, \qquad r_a + q_v \geqslant 5. \tag{5}$$

For $b > 0$, each nonempty missing row has at least $b + 2$ entries, because $q_v \leqslant 3 - b$. Since $m = b + 3$, at most one row is nonempty. Thus $s_v \leqslant 1$, and the edges of B form a matching. If $b = 1$, its endpoints have $q_v = 0$; (5) requires $r_a \geqslant 5$, although $m = 4$. If $b = 2$, the unique row misses the four matching endpoints and one isolated vertex; if $b = 3$, it misses all six vertices. In either case

$$Q \cong K_4 \vee (3K_2 \cup K_1).$$

This graph is Hamiltonian: place its three two-vertex blocks and singleton between successive vertices of K_4 in a cycle. Therefore $b = 0$, and again $L_{10} \subseteq K_4 \vee \overline{K}_6$.

3.4 Reconstruction

The core $K_2 \vee (K_5 \cup \overline{K}_3)$ is impossible. Its three independent degree-two vertices would all have to be neighbors of the first reconstructed vertex to give $\delta(L_{11}) \geqslant 3$, but that neighborhood is a triangle.

For $L_{10} = K_3 \vee (K_3 \cup \overline{K}_4)$, let A be the joining triangle, C the other triangle, and B the independent set. Every vertex outside A starts a nine-vertex path, as the following templates show:

$$c_1, c_2, c_3, a_1, b_1, a_2, b_2, a_3, b_3,$$
$$b_1, a_1, c_1, c_2, c_3, a_2, b_2, a_3, b_3.$$

A newly added vertex cannot be adjacent to such an endpoint, so its neighborhood must be exactly A. The same templates apply to it, and induction forces all seven added vertices to have neighborhood A. The resulting graph is $K_3 \vee (K_3 \cup \overline{K}_{11})$.

Finally, suppose $L_{10} \subseteq K_4 \vee \overline{K}_6$, with parts A, B. The container has 30 edges, so at most three cross edges are missing. Every vertex $z \in B$ starts a nine-vertex path, as does any added vertex adjacent to A and distinct from the original core. Choose a neighbor a_1 of z and order the other three vertices of A. Use five original vertices of B, excluding z if necessary. Find distinct representatives in these five vertices for the four slots

$$\{a_1, a_2\}, \quad \{a_2, a_3\}, \quad \{a_3, a_4\}, \quad \{a_4\},$$

where eligibility means adjacency to every vertex in the slot. Each slot has at least two eligible vertices. A vertex ineligible for every slot of a three- or four-slot collection requires at least two missing edges, because one missing edge affects at most two slots. With at most three missing edges, at most one of the five vertices can be excluded from such a union. These unions have at least four eligible vertices. Hall's condition holds for one, two, three, and four slots, giving the path

$$z, a_1, b_1, a_2, b_2, a_3, b_3, a_4, b_4.$$

Consider the first added vertex whose earlier neighborhood is not contained in A. An earlier neighbor outside A starts such a path, and prepending the new vertex gives a P_{10}. No such vertex exists, so A covers the final graph. This contradicts our standing assumption.

We have proved the classification. In its triangle-join case, delete A and one vertex of C, leaving one edge. In the cover case, deleting four vertices containing the cover leaves none. This proves Theorem 2.

4 The exact threshold and earlier certificates

Proof of Theorem 1. In G_{16}, partition the eight odd vertices into $A = \{1, 3, 5, 7\}$ and $B = \{9, 11, 13, 15\}$. Color every edge meeting A red and every other edge blue. There are no even–even edges, so A and B are covers for the respective colors. A graph with a four-vertex cover has no path with more than eight edges: a path uses at most four cover vertices and five other vertices. Thus $P_{\mathrm{cop}}(9) > 16$.

If an avoiding coloring existed on G_{17}, Proposition 5 would give a connected 48-edge color graph retaining at least five edges after every four-vertex deletion. Theorem 2 gives a deletion leaving at most one edge, a contradiction. □

The same cover argument on G_{12} gives a coloring with covers $\{1, 3, 11\}$ and $\{5, 7, 9\}$, whose two longest path lengths are six. At $n = 13$, an archived 57-variable Boolean formula contains 71 832 valid eight-edge-path exclusion clauses and one tautology registering all variables. No symmetry restrictions are imposed. Each path contributes clauses excluding its two monochromatic assignments. An independently checked DRAT proof establishes that this subset of path constraints is unsatisfiable. Thus every coloring has an eight-edge path, which also contains a seven-edge path, proving the previously reported values

$$P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13.$$

The numerical values are prior public results; the certificate archive closes their reproducibility gap. Their proof checker and trust boundary are distinct from the positive certificates used at $n = 17$.

5 Linear bounds

Proof of Theorem 3. The $2m$ odd vertices cover G_{4m}. Partition them into two sets of size m and color edges by the first set they meet, as above. Each color has a cover of size m, so its paths have at most $2m$ edges. Hence $P_{\mathrm{cop}}(2m+1) \geqslant 4m+1$. Monotonicity in the target length gives the stated finite lower bound and asymptotic lower constant two.

For an upper bound, the Erdős–Gallai path theorem [5] implies that a graph without a k-edge path has at most $(k-1)v/2$ edges. Applying this separately to both colors of any subgraph H of G_n gives the sufficient condition

$$e(H) > (k-1)v(H) \quad \Longrightarrow \quad P_{\mathrm{cop}}(k) \leqslant n. \tag{6}$$

The strict inequality is essential. Write $E_n = e(G_n)$. Counting ordered coprime pairs by Möbius inversion gives

$$2E_n + 1 = \sum_{r=1}^{n} \mu(r)\lfloor n/r \rfloor^2 = \frac{6}{\pi^2} n^2 + O(n \log n).$$

The extra one counts $(1,1)$. The floor error is $O(n \log n)$ and the tail of the absolutely convergent series $\sum \mu(r)/r^2 = 1/\zeta(2)$ contributes $O(n)$. Using $H = G_n$ in (6) already gives $\limsup P_{\mathrm{cop}}(k)/k \leqslant \pi^2/3$.

For the improved constant, delete

$$D_n = \{a \leqslant n : 30 \mid a \text{ or } 42 \mid a \text{ or } 70 \mid a\}, \qquad H_n = G_n - D_n.$$

All deleted vertices are even, so D_n is independent. Each pair of these divisors has least common multiple 210. Inclusion-exclusion gives

$$|D_n| = \lfloor n/30 \rfloor + \lfloor n/42 \rfloor + \lfloor n/70 \rfloor - 2\lfloor n/210 \rfloor = \frac{13}{210} n + O(1).$$

For fixed squarefree $d \geqslant 2$, let $C_d(n)$ count ordered coprime pairs $(a,b) \in [n]^2$ with $d \mid a$. Then

$$\begin{aligned} C_d(n) &= \sum_{r=1}^{n} \mu(r) \left\lfloor \frac{n}{\operatorname{lcm}(d,r)} \right\rfloor \left\lfloor \frac{n}{r} \right\rfloor \\ &= \frac{6n^2}{\pi^2 \prod_{p \mid d}(p+1)} + O_d(n \log n). \end{aligned}$$

Indeed, removing floors and extending the sum gives the same errors as above, and the leading coefficient is

$$\sum_{r \geqslant 1} \frac{\mu(r)}{r \operatorname{lcm}(d,r)} = \frac{1}{d} \prod_{p \mid d} (1 - 1/p) \prod_{p \nmid d} (1 - 1/p^2) = \frac{6}{\pi^2 \prod_{p \mid d}(p+1)}.$$

Independence of D_n means that edges lost are counted exactly once by pairs whose first endpoint is deleted. Their number is

$$C_{30} + C_{42} + C_{70} - 2C_{210} = \frac{6}{\pi^2}\left(\frac{1}{72} + \frac{1}{96} + \frac{1}{144} - \frac{2}{576}\right) n^2 + O(n \log n)$$
$$= \frac{n^2}{6\pi^2} + O(n \log n).$$

Thus

$$v(H_n) = \frac{197}{210} n + O(1), \qquad e(H_n) = \frac{17}{6\pi^2} n^2 + O(n \log n),$$

and $e(H_n)/v(H_n) = 595n/(197\pi^2) + O(\log n)$. For any $\varepsilon > 0$, set $n = \lceil (197\pi^2/595 + \varepsilon)k \rceil$ and use (6) for sufficiently large k. □

6 Scope and related extremal results

Theorem 2 concerns a single equality case. Its 18-vertex augmentation has 65 edges, exactly $h(18, 11, 4)$. The strict stability bound alone therefore does not settle it. In the notation of Ma and Ning [8], $W_{n,a,c} = H(n, c+1, a)$; at $(n, c) = (18, 10)$ the term $f(n, 4, c)$ is also 65. Their Theorems 1.7 and 1.9 retain a strict inequality above a maximum containing this term. This comparison explains why those statements do not directly imply our boundary classification. It is not a comprehensive priority determination.

The result $P_{\mathrm{cop}}(9) = 17$ is consistent with $P_{\mathrm{cop}}(2m + 1) = 4m + 1$, whose lower bound holds for every m. The proof here does not establish that infinite family. It also leaves a gap between the asymptotic constants two and $197\pi^2/595$. A shorter structural proof for $P_{\mathrm{cop}}(7) = P_{\mathrm{cop}}(8) = 13$ would remove the separate SAT certificate from the account of those earlier values. None of these questions is resolved by a time-limited search returning no answer.

A Evidence and reproducibility

All evidence used here is available in the repository at the immutable revision `8bda81e` [9]. Paths below are relative to its root. This revision predates the present manuscript but contains its proof notes, certificates, and checkers. The evidence is organized as follows:

Finite host lemmas. The structural archive contains certificates for the crossing paths, deletion families, and cut extensions in the three finite lemmas.

Nine-edge threshold. In the resolution archive, `lower-n16-k9.json` records all 79 colored edges of the lower witness, alongside source and verification records.

Earlier thresholds. The earlier path archive contains the 45-edge lower witness at $n = 12$, and the compressed formula and DRAT proof at $n = 13$.

The first two archives use standard-library Python checkers; no SAT solver is needed for their verification. For example, from the root, with Python 3.10 or newer:

```
python3 -I -S -O src/coprime_path_structure_certificates.py \
  cuts docs/audits/coprime-path-nine-2026-10-18/cuts.json.gz
python3 -I -S -O src/coprime_path_structure_certificates.py \
  faults docs/audits/coprime-path-nine-2026-10-18/fault-families.json.gz
python3 -I -S -O src/coprime_path_structure_certificates.py \
  extensions docs/audits/coprime-path-nine-2026-10-18/extensions.json.gz
```

Require successful exit and the case counts in Section 2; also require the archived task parameters $n = 17$, target length nine, minimum cut side five, and both deletion budgets four. For extensions, require `side_size=4` and `minimum_side_edges=5`. The complete lower colorings are independently checked by subset dynamic programming, which records reachable path endpoints for every vertex subset and color. Their longest lengths are $(8, 8)$ at $n = 16$ and $(6, 6)$ at $n = 12$.

For the $n = 13$ upper bound, the separate formula checker verifies that every non-tautological clause consists of eight distinct edges of one simple path and has uniform sign. DRAT-trim, pinned at revision `2e3b2dc0ecf938addbd779d42877b6ed69d9a985`, then verifies the unsatisfiability proof. The archive README gives complete commands. Its reported proof core has 57 506 clauses, 391 234 lemmas, and 4 858 014 resolution steps, with no RAT lemmas in the core.

The positive-certificate checker verifies host symmetries only; it never assumes that an edge coloring respects them. The trust boundary comprises the checkers, their execution environments, and the published mathematical inputs. Tests of auxiliary finite steps do not formally verify the full written proof. There is no Lean formalization of this argument, and the $n = 17$ proof does not rely on a SAT unsatisfiability certificate.

The earlier code archive at doi:10.5281/zenodo.23433640 and the earlier manuscript record at doi:10.5281/zenodo.19058647 are retained as historical records. They are not archives of the new nine-edge-path evidence; the immutable repository revision above identifies that evidence precisely.

AI assistance and draft status

Codex assisted the proof investigation, certificate software, literature checks, and preparation of this working manuscript. Automated checking and a separate AI review are not human mathematical review. Human verification of every proof step and citation, and rewriting of AI-assisted proofs in the author's own words, remain pending. This draft is not a journal submission and makes no claim that those requirements have been completed.

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